

The Composition of Linear Transformations

An Algebraic Perspective of Matrix Multiplication

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Abstract

This paper presents a rigorous study of composition of linear transformations and multiplication of their corresponding coordinate matrices. Here we establish that a matrix multiplication is the exact algebraic representation of the composition of their corresponding transformations.

1 Introduction

We know a transformation $T : V \rightarrow W$ describes how one space maps into another. The natural progression of this concept arises when we consider a second linear transformation, $U : W \rightarrow Z$. By applying U to the image of T , we establish a chained operation known as composition.

2 Composition of Linear Transformations

Composition of linear transformations is just like composite functions. Just as we show $f \circ g(x) = f(g(x))$, similarly in linear algebra we represent our composition of transformations as $UT(x)$ not $U \circ T$ and we define $UT(x) = U(T(x))$.

Theorem 2.1. *Let V, W and Z be vector spaces over the same field F , and let $T : V \rightarrow W$ and $U : W \rightarrow Z$ be linear. Then $UT : V \rightarrow Z$ is linear.*

Proof. Since T and U are linear, and let $x, y \in V$ and $a \in F$. Then

$$\begin{aligned} UT(ax + y) &= U(T(ax + y)) = U(aT(x) + T(y)) \\ &= aU(T(x)) + U(T(y)) = a(UT)(x) + (UT)(y) \end{aligned}$$

Hence, UT is linear □

Theorem 2.2. *Let V be a vector space. Let $T, U_1, U_2 \in \mathcal{L}(V)$. Then*

1. $T(U_1 + U_2) = TU_1 + TU_2$ and $(U_1 + U_2)T = U_1T + U_2T$.
2. $T(U_1U_2) = (TU_1)U_2$.
3. $TI = IT = T$.

4. $a(U_1U_2) = (aU_1)U_2 = U_1(aU_2).$

Proof. Given, $T, U_1, U_2 \in \mathcal{L}(V)$. Then T, U_1 & U_2 are linear. Then for some $v \in V$, we have

1.

$$\begin{aligned}(T(U_1 + U_2))(v) &= T((U_1 + U_2)(v)) = T(U_1(v) + U_2(v)) \\ &= T(U_1(v)) + T(U_2(v)) = TU_1(v) + TU_2(v)\end{aligned}$$

Also

$$\begin{aligned}((U_1 + U_2)T)(v) &= (U_1 + U_2)(T(v)) = U_1(T(v)) + U_2(T(v)) \\ &= U_1T(v) + U_2T(v)\end{aligned}$$

2.

$$(T(U_1U_2))(v) = T((U_1U_2)(v)) = T(U_1(U_2(v))) = (TU_1)(U_2(v))$$

3.

$$(TI)(v) = T(I(v)) = T(v)$$

Also,

$$(IT)(v) = I(T(v)) = T(v)$$

4.

$$(a(U_1U_2))(v) = a((U_1U_2)(v)) = a(U_1(U_2(v)))$$

Now

$$((aU_1)U_2)(v) = (aU_1)(U_2(v)) = a(U_1(U_2(v)))$$

Thus, $a(U_1U_2) = (aU_1)U_2$.

Also

$$(U_1(aU_2))(v) = U_1((aU_2)(v)) = U_1(a(U_2(v))) = a(U_1(U_2(v)))$$

Thus, $a(U_1U_2) = U_1(aU_2)$

□

3 Matrix Multiplication

Let $T : V \rightarrow W$ and $U : W \rightarrow Z$ be linear transformations, and let $A = [U]_\beta^\gamma$ and $B = [T]_\alpha^\beta$ where $\alpha = \{v_1, v_2, \dots, v_n\}$ and $\beta = \{w_1, w_2, \dots, w_m\}$ and $\gamma = \{z_1, z_2, \dots, z_p\}$ are ordered bases for V, W and Z , respectively. Then the product of two matrices are defined as $AB = [UT]_\alpha^\gamma$.

For $1 \leq j \leq n$, we have

$$\begin{aligned}
UT(v_j) &= U(T(v_j)) = U\left(\sum_{k=1}^m B_{kj}w_k\right) = \sum_{k=1}^m B_{kj}w_k \\
&= \sum_{k=1}^m B_{kj}\left(\sum_{i=1}^p A_{ik}z_i\right) = \sum_{i=1}^p \left(\sum_{k=1}^m A_{ik}B_{kj}\right) z_i \\
&= \sum_{i=1}^p C_{ij}z_i
\end{aligned}$$

where

$$C_{ij} = \sum_{k=1}^m A_{ik}B_{kj}$$

Definition 3.1. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. We define the product of A and B , denoted AB , to be the $m \times p$ matrix such that

$$(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj} \quad \text{for } 1 \leq i \leq m, 1 \leq j \leq p$$

Theorem 3.1. Let V, W and Z be finite dimensional vector spaces with ordered bases α, β and γ , respectively. Let $T : V \rightarrow W$ and $U : W \rightarrow Z$ be linear. Then,

$$[UT]_{\alpha}^{\gamma} = [U]_{\beta}^{\gamma}[T]_{\alpha}^{\beta}$$

Proof. Using the above notation, we have,

$$[UT]_{\alpha}^{\gamma} = AB$$

where $A = [U]_{\beta}^{\gamma}$ and $B = [T]_{\alpha}^{\beta}$.

Thus,

$$[UT]_{\alpha}^{\gamma} = AB = [U]_{\beta}^{\gamma}[T]_{\alpha}^{\beta}$$

□

Corollary 3.1.1. Let V be a finite dimensional vector space with an ordered basis β . Let $T, U \in \mathcal{L}(V)$. Then

$$[UT]_{\beta} = [U]_{\beta}[T]_{\beta}$$

Proof. This follows directly from previous theorem if $V = W = Z$ and $\alpha = \beta = \gamma$. Then,

$$[UT]_{\beta} = [U]_{\beta}[T]_{\beta}$$

□

Definition 3.2. We define the Kronecker delta δ_{ij} by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. The $n \times n$ identity matrix I_n is defined by $(I_n)_{ij} = \delta_{ij}$

Theorem 3.2. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and D and E be $q \times m$ matrices. Then

1. $A(B + C) = AB + AC$ and $(D + E)A = DA + EA$
2. $a(AB) = (aA)B = A(aB)$ for any scalar a .
3. $I_m A = A = A I_n$
4. If V is an n -dimensional vector space with an ordered basis β , then $[I_V]_\beta = I_n$.

Proof. 1.

$$\begin{aligned}
[A(B + C)]_{ij} &= \sum_{k=1}^n A_{ik}(B + C)_{kj} = \sum_{k=1}^n A_{ik}(B_{kj} + C_{kj}) \\
&= \sum_{k=1}^n (A_{ik}B_{kj} + A_{ik}C_{kj}) = \sum_{k=1}^n A_{ik}B_{kj} + \sum_{k=1}^n A_{ik}C_{kj} \\
&= (AB)_{ij} + (AC)_{ij} = [AB + AC]_{ij}
\end{aligned}$$

$$\therefore A(B + C) = AB + AC$$

Again,

$$\begin{aligned}
[(D + E)A]_{ij} &= \sum_{k=1}^m (D + E)_{ik}A_{kj} = \sum_{k=1}^m (D_{ik} + E_{ik})A_{kj} \\
&= \sum_{k=1}^m (D_{ik}A_{kj} + E_{ik}A_{kj}) = (DA)_{ij} + (EA)_{ij} \\
&= [DA + EA]_{ij}
\end{aligned}$$

$$\therefore (D + E)A = DA + EA$$

2. We have,

$$\begin{aligned}
[a(AB)]_{ij} &= a[AB]_{ij} = a \sum_{k=1}^n A_{ik}B_{kj} = \sum_{k=1}^n a(A_{ik}B_{kj}) \\
&= \sum_{k=1}^n (aA)_{ik}B_{kj} = [(aA)B]_{ij} \\
\text{or} \quad &= \sum_{k=1}^n A_{ik}(aB)_{kj} = [A(aB)]_{ij}
\end{aligned}$$

So,

$$a(AB) = (aA)B = A(aB)$$

3. We have,

$$[I_m A]_{ij} = \sum_{k=1}^m (I_m)_{ik}A_{kj} = \sum_{k=1}^m \delta_{ik}A_{kj} = A_{ij}$$

Again,

$$[A I_n]_{ij} = \sum_{k=1}^n A_{ik}(I_n)_{kj} = \sum_{k=1}^n A_{ik}\delta_{kj} = A_{ij}$$

So

$$I_m A = A = A I_n$$

4. $I_V(v_j) = v_j$. Thus,

$$I_V(v_j) = 0v_1 + 0v_2 + \cdots + 1v_j + \cdots + 0v_n$$

or we can write,

$$I_V(v_j) = \sum_{i=1}^n \delta_{ij} v_i$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$, which is the definition of Kronecker delta. Thus, $[I_V]_\beta = I_n$. □

Corollary 3.2.1. *Let A be an $m \times n$ matrix and $B_1, B_2, \dots, B_n$ be $n \times p$ matrices and $C_1, C_2, \dots, C_p$ be $q \times m$ matrices, and $a_1, a_2, \dots, a_k$ be scalars. Then*

$$A \left(\sum_{i=1}^k a_i B_i \right) = \sum_{i=1}^k a_i (AB_i) \quad \text{and} \quad \left(\sum_{i=1}^k a_i C_i \right) A = \sum_{i=1}^k a_i (C_i A)$$

Proof. Proving $A \left(\sum_{i=1}^k a_i B_i \right) = \sum_{i=1}^k a_i (AB_i)$:

Let $k = 1$ then, $A(a_1 B_1) = a_1 (AB_1)$ by Theorem-3.2(2).

Let $k = 2$ then, $A(a_1 B_1 + a_2 B_2) = a_1 (AB_1) + a_2 (AB_2)$ by Theorem-3.2(1),(2).

Thus the theorem is true for $k = 1, 2$. Now, let the equation be true for an arbitrary $k = m$. Then,

$$\begin{aligned} & A \left(\sum_{i=1}^m a_i B_i \right) = \sum_{i=1}^m a_i (AB_i) \\ \implies & A \left(\sum_{i=1}^m a_i B_i + a_{m+1} B_{m+1} - a_{m+1} B_{m+1} \right) = \sum_{i=1}^m a_i (AB_i) \\ \implies & A \left(\sum_{i=1}^{m+1} a_i B_i \right) - a_{m+1} (AB_{m+1}) = \sum_{i=1}^m a_i (AB_i) \\ \implies & A \left(\sum_{i=1}^{m+1} a_i B_i \right) = \sum_{i=1}^m a_i (AB_i) + a_{m+1} (AB_{m+1}) \\ & = \sum_{i=1}^{m+1} a_i (AB_i) \end{aligned}$$

Thus the theorem is true for any k by the method of induction.

Proving $\left(\sum_{i=1}^k a_i C_i \right) A = \sum_{i=1}^k a_i (C_i A)$:

Let $k = 1$ then, $(a_1 C_1) A = a_1 (C_1 A)$ by Theorem-3.2(2).

Let $k = 2$ then, $(a_1 C_1 + a_2 C_2) A = a_1 (C_1 A) + a_2 (C_2 A)$ by Theorem-3.2(1),(2).

Thus the theorem is true for $k = 1, 2$. Now, let the equation be true for an arbitrary $k = m$. Then,

$$\begin{aligned}
& \left(\sum_{i=1}^m a_i C_i \right) A = \sum_{i=1}^m a_i (C_i A) \\
\Rightarrow & \left(\sum_{i=1}^m a_i C_i + a_{m+1} C_{m+1} - a_{m+1} C_{m+1} \right) A = \sum_{i=1}^m a_i (C_i A) \\
& \Rightarrow \left(\sum_{i=1}^{m+1} a_i C_i \right) A - a_{m+1} (C_{m+1} A) = \sum_{i=1}^m a_i (C_i A) \\
& \Rightarrow \left(\sum_{i=1}^{m+1} a_i C_i \right) A = \sum_{i=1}^m a_i (C_i A) + a_{m+1} (C_{m+1} A) \\
& = \sum_{i=1}^{m+1} a_i (C_i A)
\end{aligned}$$

Thus the theorem is true for any k by the method of induction. $\square$

Theorem 3.3. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. For each j ($1 \leq j \leq p$) let u_j and v_j denote the j -th columns of AB and B , respectively. Then

1. $u_j = Av_j$
2. $v_j = Be_j$, where e_j is the j -th standard vector of F^p

Proof. Given,

$$u_j = \begin{bmatrix} (AB)_{1j} \\ (AB)_{2j} \\ \vdots \\ (AB)_{mj} \end{bmatrix} \quad \text{and} \quad v_j = \begin{bmatrix} B_{1j} \\ B_{2j} \\ \vdots \\ B_{nj} \end{bmatrix}$$

1. We have

$$\begin{aligned}
u_j &= \begin{bmatrix} (AB)_{1j} \\ (AB)_{2j} \\ \vdots \\ (AB)_{mj} \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^n A_{1k} B_{kj} \\ \sum_{k=1}^n A_{2k} B_{kj} \\ \vdots \\ \sum_{k=1}^n A_{mk} B_{kj} \end{bmatrix} = \begin{bmatrix} A_{11} B_{1j} + A_{12} B_{2j} + \dots + A_{1n} B_{nj} \\ A_{21} B_{1j} + A_{22} B_{2j} + \dots + A_{2n} B_{nj} \\ \vdots \\ A_{m1} B_{1j} + A_{m2} B_{2j} + \dots + A_{mn} B_{nj} \end{bmatrix} \\
&= \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix} \cdot \begin{bmatrix} B_{1j} \\ B_{2j} \\ \vdots \\ B_{nj} \end{bmatrix} = Av_j
\end{aligned}$$

2. Since, e_j is the j -th standard vector of F^p , we have

$$e_j = (0, 0, \dots, 0, 1, 0, \dots, 0)$$

Again

$$B_{ij} = (BI_n)_{ij} = \sum_{k=1}^p B_{ik}\delta_{kj}$$

Here,

$$(e_j)_i = \delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$$\therefore B_{ij} = \sum_{k=1}^p B_{ik}(e_j)_k = (Be_j)_i$$

$$Be_j = \begin{bmatrix} B_{11}(0) + B_{12}(0) + \dots + B_{1j}(1) + \dots + B_{1p}(0) \\ B_{21}(0) + B_{22}(0) + \dots + B_{2j}(1) + \dots + B_{2p}(0) \\ \vdots \\ B_{n1}(0) + B_{n2}(0) + \dots + B_{nj}(1) + \dots + B_{np}(0) \end{bmatrix} = \begin{bmatrix} B_{1j} \\ B_{2j} \\ \vdots \\ B_{nj} \end{bmatrix} = v_j$$

□

Theorem 3.4. Assuming the notation of the previous theorem.

1. Suppose the z is a column vector in F^p then Bz is a linear combination of columns of B , i.e., if $z = (a_1, a_2, \dots, a_p)^t$ then

$$Bz = \sum_{j=1}^p a_j v_j$$

2. Extend (1) to column j of AB is a linear combination of the columns of A with the coefficients in the linear combination being the entries of column j of B .
3. For any row vector $w \in F^m$, prove that wA is a linear combination of the rows of A with the coefficients in the linear combination being the coordinates of w .
4. Extend (3) to row i of AB is a linear combination of the rows of B with the coefficients in the linear combination being the entries of row i of A .

Proof. From previous theorem, we have, A is an $m \times n$ matrix and B is an $n \times p$ matrix. If $1 \leq j \leq p$ then

$$u_j = \begin{bmatrix} (AB)_{1j} \\ (AB)_{2j} \\ \vdots \\ (AB)_{mj} \end{bmatrix} \quad \text{and} \quad v_j = \begin{bmatrix} B_{1j} \\ B_{2j} \\ \vdots \\ B_{nj} \end{bmatrix}$$

Also, $u_j = Av_j$ and $v_j = Be_j$, where e_j is standard j -th vector of F^p .

1.

$$B = \begin{bmatrix} B_{11} & B_{12} & \dots & B_{1p} \\ B_{21} & B_{22} & \dots & B_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \dots & B_{np} \end{bmatrix} \quad \text{and} \quad z = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix}$$

Then

$$Bz = \begin{bmatrix} a_1B_{11} + a_2B_{12} + \dots + a_pB_{1p} \\ a_1B_{21} + a_2B_{22} + \dots + a_pB_{2p} \\ \vdots \\ a_1B_{n1} + a_2B_{n2} + \dots + a_pB_{np} \end{bmatrix}$$

By matrix addition and scalar multiplication of matrices, we have

$$\begin{aligned} Bz &= \begin{bmatrix} a_1B_{11} \\ a_1B_{21} \\ \vdots \\ a_1B_{n1} \end{bmatrix} + \begin{bmatrix} a_2B_{12} \\ a_2B_{22} \\ \vdots \\ a_2B_{n2} \end{bmatrix} + \dots + \begin{bmatrix} a_pB_{1p} \\ a_pB_{2p} \\ \vdots \\ a_pB_{np} \end{bmatrix} \\ &= a_1 \begin{bmatrix} B_{11} \\ B_{21} \\ \vdots \\ B_{n1} \end{bmatrix} + a_2 \begin{bmatrix} B_{12} \\ B_{22} \\ \vdots \\ B_{n2} \end{bmatrix} + \dots + a_p \begin{bmatrix} B_{1p} \\ B_{2p} \\ \vdots \\ B_{np} \end{bmatrix} \\ &= a_1v_1 + a_2v_2 + \dots + a_pv_p \\ &= \sum_{j=1}^p a_jv_j \end{aligned}$$

Thus we prove that if we multiply a vector to a matrix then the entries of the vector acts as coefficients to the linear combination of the columns of the matrix.

2. Let $w_1, a_2, \dots, a_n$ be the columns of matrix A , *i.e.*,

$$a_j = \begin{bmatrix} A_{1j} \\ A_{2j} \\ \vdots \\ A_{mj} \end{bmatrix} \quad 1 \leq j \leq n.$$

Also, we have v_j to be the j -th column of B and also that $u_j = Av_j$. Now from part (1), we have if we multiply a matrix with a vector then the entries of vector act as coefficients in the linear combination of the columns of matrix. Thus,

$$u_j = B_{1j}a_1 + B_{2j}a_2 + \dots + B_{nj}a_n = \sum_{k=1}^n B_{kj}a_k$$

Thus we prove that j -th column of AB is a linear combination of columns of A with j -th column of B as the coefficients.

3. We have w is a row vector in F^m then we can write w as

$$w = \begin{bmatrix} w_1 & w_2 & \dots & w_m \end{bmatrix}$$

Also we have A is an $m \times n$ matrix. Then wA is defined and,

$$\begin{aligned}
wA &= \begin{bmatrix} w_1 & w_2 & \dots & w_m \end{bmatrix} \cdot \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix} \\
&= \begin{bmatrix} w_1A_{11} + w_2A_{21} + \dots + w_mA_{m1} & w_1A_{12} + w_2A_{22} + \dots + w_mA_{m2} & \dots & w_1A_{1n} + w_2A_{2n} + \dots + w_mA_{mn} \end{bmatrix} \\
&= \begin{bmatrix} w_1A_{11} & w_1A_{12} & \dots & w_1A_{1n} \end{bmatrix} + \begin{bmatrix} w_2A_{21} & w_2A_{22} & \dots & w_2A_{2n} \end{bmatrix} \\
&\quad \dots + \begin{bmatrix} w_mA_{m1} & w_mA_{m2} & \dots & w_mA_{mn} \end{bmatrix} \\
&= w_1 \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \end{bmatrix} + w_2 \begin{bmatrix} A_{21} & A_{22} & \dots & A_{2n} \end{bmatrix} \\
&\quad \dots + w_m \begin{bmatrix} A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}
\end{aligned}$$

If we take $A_1, A_2, \dots, A_m$ to be the rows of A , then

$$wA = w_1A_1 + w_2A_2 + \dots + w_mA_m = \sum_{k=1}^m w_k A_k$$

Thus we prove that wA is a linear combination of the rows of A with coefficients being the coordinates of w .

4. let A_i and B_j be the rows of A and B respectively, such that

$$A_i = \begin{bmatrix} A_{i1} & A_{i2} & \dots & A_{in} \end{bmatrix} \quad \text{and} \quad B_j = \begin{bmatrix} B_{j1} & B_{j2} & \dots & B_{jp} \end{bmatrix}$$

for $1 \leq i \leq m$ and $1 \leq j \leq n$, respectively. Here

$$\begin{aligned}
AB &= \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \end{bmatrix} B + \begin{bmatrix} A_{21} & A_{22} & \dots & A_{2n} \end{bmatrix} B \\
&\quad \dots + \begin{bmatrix} A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix} B
\end{aligned}$$

Since, in part (3), we proved that if we multiply a row vector with a matrix then it forms a linear combination of rows of matrix then it forms a linear combination of rows of matrix with the coefficients being the coordinates of row vector. Then,

$$AB = \begin{bmatrix} \sum_{k=1}^n A_{1k}B_k \\ \sum_{k=1}^n A_{2k}B_k \\ \vdots \\ \sum_{k=1}^n A_{mk}B_k \end{bmatrix} \begin{matrix} \rightarrow \text{1st row} \\ \rightarrow \text{2nd row} \\ \\ \rightarrow \text{m-th row} \end{matrix}$$

Thus

$$\text{i-th row of } AB = \sum_{k=1}^n A_{ik} B_k$$

Hence we proved that i-th row of AB is a linear combination of rows of B with coefficients being the entries of i-th rows of A .

□

Theorem 3.5. *Let V and W be finite dimensional vector spaces having ordered bases β and γ , respectively, and let $T : V \rightarrow W$ be linear. Then for each $u \in V$, we have*

$$[T(u)]_\gamma = [T]_\beta^\gamma [u]_\beta$$

Proof. Let $\beta = \{v_1, v_2, \dots, v_n\}$ and $\gamma = \{w_1, w_2, \dots, w_m\}$ be the bases of V and W respectively. Since $u \in V$ then,

$$u = \sum_{j=1}^n c_j v_j \text{ for } c_j \in F$$

Then,

$$[u]_\beta = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Applying transformation on u ,

$$T(u) = T\left(\sum_{j=1}^n c_j v_j\right) = \sum_{j=1}^n c_j T(v_j) \quad (1)$$

Also by the definition of matrix representation we can write,

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i \quad \text{for } 1 \leq j \leq n.$$

Then,

$$[T]_\beta^\gamma = A$$

Again,

$$\begin{aligned} (1) \implies T(u) &= \sum_{j=1}^n c_j T(v_j) = \sum_{j=1}^n c_j \left(\sum_{i=1}^m A_{ij} w_i \right) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} c_j \right) w_i \end{aligned}$$

$$\therefore T(u) = \left(\sum_{j=1}^n A_{1j} c_j \right) w_1 + \left(\sum_{j=1}^n A_{2j} c_j \right) w_2 + \dots + \left(\sum_{j=1}^n A_{mj} c_j \right) w_m$$

Since, $w_i \in \gamma$. Then,

$$[T(u)]_\gamma = \begin{bmatrix} \sum_{j=1}^n A_{1j}c_j \\ \sum_{j=1}^n A_{2j}c_j \\ \vdots \\ \sum_{j=1}^n A_{mj}c_j \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = A \cdot [u]_\beta = [T]_\beta^\gamma [u]_\beta$$

$$\therefore [T(u)]_\gamma = [T]_\beta^\gamma [u]_\beta$$

□

Definition 3.3. Let A be an $m \times n$ matrix with entries from field F . We denote by L_A the mapping $L_A : F^n \rightarrow F^m$ defined by $L_A(x) = Ax$ (the matrix product of A and x) for each column vector $x \in F^n$. We call L_A a left-multiplication transformation.

Theorem 3.6. Let A be an $m \times n$ matrix with entries from F . Then $L_A : F^n \rightarrow F^m$ is linear. Also if B is any other $m \times n$ matrix (with entries from F) and β and γ are the standard ordered bases for F^n and F^m , respectively, then the following is true:

1. $[L_A]_\beta^\gamma = A$
2. $L_A = L_B$ if and only if $A = B$
3. $L_{A+B} = L_A + L_B$ and $L_{aA} = aL_A$ for all $a \in F$.
4. If $T : F^n \rightarrow F^m$ is linear, then there exists a unique $m \times n$ matrix C such that $T = L_C$. In fact, $C = [T]_\beta^\gamma$.
5. If E is an $n \times p$ matrix, then $L_{AE} = L_A \cdot L_E$.
6. If $m = n$ then, $L_{I_n} = I_{F^n}$

Proof. 1. The j -th column of $[L_A]_\beta^\gamma$ is equal to $L_A(e_j)$. However $L_A(e_j) = Ae_j$, which is also the j -th column of A , by Theorem 3.3(2). Thus, $[L_A]_\beta^\gamma = A$.

2. If $L_A = L_B$, from part (1), $A = [L_A]_\beta^\gamma = [L_B]_\beta^\gamma = B$. Also if $A = B$ then, $L_A(x) = Ax = Bx = L_B(x)$ for all $x \in F^n$.

3. $L_{A+B}(x) = (A+B)x = Ax + Bx = L_A(x) + L_B(x)$ for $x \in F^n$.

Again, $L_{aA}(x) = (aA)x = a(Ax) = aL_A(x)$ for all $a \in F$ and $x \in F^n$.

4. Let $C = [T]_\beta^\gamma$. By Theorem 3.5, $[T(x)]_\gamma = [T]_\beta^\gamma [x]_\beta$, or $T(x) = Cx = L_C(x)$ for all $x \in F^n$. So $T = L_C$. The uniqueness of C follows from (2).

5. Since A is an $m \times n$ matrix and E is an $n \times p$ matrix then AE is a matrix of order $m \times p$. For any j ($1 \leq j \leq p$), we can say that $(AE)e_j$ is the j -th column of AE which is also equal to $A(Ee_j)$. Thus, $(AE)e_j = A(Ee_j)$. Thus,

$$L_{AE}(e_j) = (AE)e_j = A(Ee_j) = A \cdot L_E(e_j) = L_A(L_E(e_j)).$$

Hence, $L_{AE} = L_A \cdot L_E$.

6. For any $x \in F^n$, $L_{I_n}(x) = I_n x = x$ and again, $I_{F^n}(x) = x$. Thus, $L_{I_n}(x) = x = I_{F^n}(x)$. Thus,

$$L_{I_n} = I_{F^n}$$

□

Theorem 3.7. *Let A, B and C be matrices such that $A(BC)$ is defined. Then $(AB)C$ is also defined and $A(BC) = (AB)C$; i.e., matrix multiplication is associative.*

Proof. Let B be an $m \times n$ matrix. We have $A(BC)$ is defined. Then BC is defined. Thus C must have the no. of rows equal to the no. of columns of B . C must have n rows. Let us say C is an $n \times p$ matrix. Again A and BC must be defined. Here BC must be an $m \times p$ matrix and thus A must have m columns. Let A be an $l \times m$ matrix.

Thus so far we have:

- A is an $l \times m$ matrix,
- B is an $m \times n$ matrix,
- C is an $n \times p$ matrix.

Thus, AB must be an $l \times n$ matrix. Thus AB is defined and C has the exact no. of rows to that of columns of AB . Thus $(AB)C$ is defined and must be a matrix of order $l \times p$.

Using Theorem 3.6 (5) and Theorem 2.2 (2) we get,

$$L_{A(BC)} = L_A \cdot L_{BC} = L_A(L_B L_C) = (L_A L_B) L_C = L_{AB} L_C = L_{(AB)C}$$

So from Theorem 3.6 (2), we get:

$$A(BC) = (AB)C$$

□

4 Conclusion

Thus throughout the study, we establish matrix multiplication aren't simple computational rules but the relationship between composition of transformations and matrix multiplication forms one of the core foundations of linear algebra. By proving that the coordinate matrix of a composed transformation UT is exactly the product of their individual matrices, $[U]_\beta^\gamma [T]_\alpha^\beta$, we bridge the abstract linear transformation vector spaces with the algebra of matrices.

References

- [1] Friedberg, S. H., Insel, A. J., & Spence, L. E. (2018). *Linear Algebra* (5th ed.). Pearson.